JEE Mathematics: Quick Review with Grading and Practice Map - Function (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (77 questions; a question that appears in two of these is counted once): High = 7 or more questions, Medium = 1 to 6 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Definition of Function

Question 01

Define function.

Model Answer

A function f from A to B is a rule that associates every element of A with exactly one element of B (no element of A left out, none with two images).

Ex: $f : \{-1, 0, 1\} \to \{0, 1, 2\}$, $f(x) = x^2$ gives $f = \{(-1, 1), (0, 0), (1, 1)\}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

How is a function from A to B represented symbolically?

Model Answer

$f : A \to B$ (read: f maps A to B); $b = f(a)$ or $f : a \mapsto b$. It can also be shown as a set of ordered pairs, an arrow diagram, a table or a graph.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 03

What do you mean by pre-image and image of an element in a function?

Model Answer

If $f(a) = b$: b is the image (f-image, value of f at a) and a is a pre-image of b. An element has exactly one image but may have many pre-images.

Ex: $f(x) = x^2$: image of -2 is 4; pre-images of 4 are ± 2.

Grade - Low

Practice map, main concept: none

Also used in: none

Domain, Codomain and Range of a Function

Question 04
What is domain of a function?

Model Answer
For $f : A \to B$ the domain is A. If only the rule is given, the domain is the set of all real x for which $f(x)$ is real and defined.
Ex: $f(x) = \sqrt{x-1}$: domain $[1, \infty)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 05
What is range of a function?

Model Answer
Range $= \{f(a) : a \in A\}$, the set of all images. For a continuous function on an interval it runs from the minimum to the maximum value.
Ex: $f(x) = x^2,\ x \in R$: range $[0, \infty)$.

Grade - Medium
Practice map, main concept: DPP: 2(7); 4(8-9)
Also used in: Sheet: J-I 12, 16; J-II 3, 9; C-II 18 | DPP: 2(10) | Question Bank: 49

Question 06
What is co-domain of a function?

Model Answer
For $f : A \to B$ the set B is the co-domain: the set in which the images are allowed to lie.
Ex: $f : R \to R$, $f(x) = x^2$: co-domain R, range $[0, \infty)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 07
What is relation between range and co-domain of a function?

Model Answer
Range $\subseteq$ co-domain. Range = co-domain exactly when f is onto.

Grade - Low
Practice map, main concept: none
Also used in: none

Types of a Function

Question 08
What do you mean by uniform and non-uniform function?

Model Answer
Uniform: one rule on the whole domain, e.g. $f(x) = x^2$. Non-uniform (piecewise): different rules on different parts of the domain.
Ex: $f(x) = x^2$ for $x < 0$ and $f(x) = x + 1$ for $x \geq 0$ is non-uniform.

Grade - Low
Practice map, main concept: none
Also used in: none

Graphical Representation of a Function

Question 09

How to check whether a graph represents a function?

Model Answer

Vertical line test: if some vertical line (parallel to the y-axis) cuts the curve at more than one point, it is not a function; if every vertical line cuts it at most once, it is a function.

Ex: the circle $x^2 + y^2 = 1$ is not a function; $y = \sqrt{1 - x^2}$ is.

Grade - Low

Practice map, main concept: none

Also used in: none

Algebra of Functions

Question 10

How to represent addition, subtraction, product and quotient of two functions?

Model Answer

$(f \pm g)(x) = f(x) \pm g(x)$ and $(fg)(x) = f(x)g(x)$ with domain $D_f \cap D_g$; $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ with domain $D_f \cap D_g - \{x : g(x) = 0\}$.

Grade - Low

Practice map, main concept: Sheet: J-I 19

Also used in: none

Question 11

How to compute domain of a function?

Model Answer

Write every condition and intersect: denominator $\neq 0$; even root: radicand ≥ 0; $\log_b a$: $a > 0$, $b > 0$, $b \neq 1$; $\sin^{-1} u, \cos^{-1} u$: $-1 \leq u \leq 1$; ${}^nC_r$: $n \in N$, $r \in W$, $r \leq n$. Do not simplify the rule first.

Ex: $\sqrt{\frac{x-3}{x+2}}$: $x \in (-\infty, -2) \cup [3, \infty)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 3-6, 18; J-II 2-3, 26, 41; J-M 1, 25-26; C-II 16, 23 | DPP: 1(3-4, 6-10); 2(2, 10) | Question Bank: 37, 39

Also used in: Sheet: J-II 6, 9, 39 | DPP: 1(5); 4(1)

Types of Standard Functions

Question 12

Plot graph of all six trigonometric functions.

Model Answer

$\sin x$, $\cos x$: waves between -1 and 1, period 2π ($\sin$ through O, $\cos$ through $(0, 1)$). $\tan x$: increasing branches, asymptotes $x = (2n+1)\frac{\pi}{2}$; $\cot x$: decreasing branches, asymptotes $x = n\pi$. $\operatorname{cosec} x$, $\sec x$: U-shaped branches outside the strip $-1 < y < 1$, touching $y = \pm 1$ where $\sin x$ or $\cos x = \pm 1$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 2

Question 13
Write domain and range of all six trigonometric functions.

Model Answer
$\sin, \cos$: R, $[-1, 1]$. tan: $R - \{(2n+1)\frac{\pi}{2}\}$, R. cot: $R - \{n\pi\}$, R. cosec: $R - \{n\pi\}$, $(-\infty, -1] \cup [1, \infty)$. sec: $R - \{(2n+1)\frac{\pi}{2}\}$, $(-\infty, -1] \cup [1, \infty)$.
Ex: domain of $\tan 2x$ is $R - \left\{(2n+1)\frac{\pi}{4}\right\}$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 1(6)

Question 14
Plot graph of exponential function.

Model Answer
$y = a^x$: passes through $(0, 1)$, lies above the x-axis, x-axis is the asymptote; increasing if $a > 1$, decreasing if $0 < a < 1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 15
What is domain and range of exponential function?

Model Answer
$f(x) = a^x$ $(a > 0,\ a \neq 1)$: domain R, range $(0, \infty)$.
Ex: $\dfrac{1}{3^x - 9}$: domain $R - \{2\}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 9

Question 16
How to solve exponential equation?

Model Answer
Same base: $a^{f(x)} = a^{g(x)} \Rightarrow f(x) = g(x)$ $(a > 0,\ a \neq 1)$. Otherwise put $t = a^x$ $(t > 0)$, solve in t and reject $t \leq 0$.
Ex: $4^x - 3 \cdot 2^x + 2 = 0 \Rightarrow t^2 - 3t + 2 = 0 \Rightarrow t = 1, 2 \Rightarrow x = 0, 1$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 3

Question 17
Plot graph of logarithmic function.

Model Answer
$y = \log_a x$ is the mirror image of $y = a^x$ in $y = x$: passes through $(1, 0)$, y-axis is the asymptote; increasing if $a > 1$, decreasing if $0 < a < 1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 18
What is domain and range of logarithmic function?

Model Answer
$\log_a x$: domain $(0, \infty)$, range R. For $\log_{g(x)} f(x)$: $f(x) > 0$, $g(x) > 0$, $g(x) \neq 1$.
Ex: $\log_x(3 - x)$: domain $(0, 1) \cup (1, 3)$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 4; J-II 2 | DPP: 1(3-4, 7-10) | Question Bank: 39

Question 19
How to solve logarithmic equation?

Model Answer
Use the laws of logs to reach $\log_a f(x) = \log_a g(x) \Rightarrow f(x) = g(x)$, or $\log_a f(x) = k \Rightarrow f(x) = a^k$; keep only roots that lie in the domain.
Ex: $\log_2 x + \log_2(x - 2) = 3 \Rightarrow x(x - 2) = 8 \Rightarrow x = 4$ ($x = -2$ rejected).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 20
What is a polynomial function?

Model Answer

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n$$

Here n is a non-negative integer, a_i are real and $a_0 \neq 0$ (all powers of x are non-negative integers).

Grade - Medium
Practice map, main concept: Sheet: J-M 27
Also used in: none

Question 21
What is degree, leading coefficient of a polynomial function?

Model Answer
Degree = highest power n; leading coefficient = a_0 (coefficient of x^n); a_n is the constant term.
Ex: $3x^4 - x + 7$: degree 4, leading coefficient 3, constant term 7.

Grade - Medium
Practice map, main concept: Sheet: J-A 12
Also used in: none

Question 22
What is an odd linear polynomial?

Model Answer
A degree-one polynomial with no constant term: $f(x) = ax$, $a \neq 0$ (an odd function).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 23
What is domain of a polynomial function?

Model Answer
Domain of every polynomial function is R.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 24
What is range of an odd degree polynomial?

Model Answer
Odd degree: range R. Even degree: range is never R (it is $[m, \infty)$ or $(-\infty, M]$).
Ex: $x^5 + x^2 + 3$ has range R; $x^8 + x^6 + 4x^2 + 7$ has range $[7, \infty)$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 25
What can you say about number of zero(es) of an odd degree polynomial?

Model Answer
An odd degree polynomial has at least one real zero (it goes from $-\infty$ to ∞); a degree-n polynomial has at most n real zeros.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 26
What is the limit of y in a polynomial when x approaches ∞ / $-\infty$?

Model Answer
$y \approx a_0 x^n$ for large $|x|$: even n: $y \to +\infty$ at both ends if $a_0 > 0$ ($-\infty$ if $a_0 < 0$); odd n: $y \to \pm\infty$ with opposite signs at the two ends.
Ex: $x^4 - 3x + 2 \to \infty$ as $x \to \pm\infty$; $x^5 + x^2 + 3 \to \pm\infty$ as $x \to \pm\infty$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 27
What is a rational function?

Model Answer
$f(x) = \dfrac{g(x)}{h(x)}$ with g, h polynomials, $h(x) \neq 0$; domain $R - \{x : h(x) = 0\}$.
Ex: $\dfrac{x^2 - 4}{x - 2}$ has domain $R - \{2\}$ (not R).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 28
What is an algebraic function?

Model Answer
A function built from polynomials by $+, -, \times, \div$ and taking roots. Every polynomial is algebraic, not conversely.
Ex: $\sqrt{x^2+1}$, $\dfrac{x^4-16x^2}{x+\sqrt{x}}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 29
What is a transcendental function?

Model Answer
A function that is not algebraic: trigonometric, inverse trigonometric, exponential, logarithmic functions.
Ex: $\sin x$, e^x, $\ln x$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 30
Plot graph of modulus function.

Model Answer

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Graph: the V-shape $y = x$ $(x \geq 0)$ and $y = -x$ $(x < 0)$, vertex at O.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 3, 6; J-M 26 | Question Bank: 37, 67

Question 31
What is domain and range of modulus function?

Model Answer
$|x|$: domain R, range $[0, \infty)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 32
How to plot the graph of $y = |f(x)|$ from the graph of $y = f(x)$?

Model Answer
Keep the part where $f(x) \geq 0$; reflect the part below the x-axis in the x-axis.
Ex: $y = |\sin x|$ (period π), $y = |\ln x|$, $y = |(x-1)(x-2)|$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 1; J-II 28, 40

Question 33
Plot graph of signum function.

Model Answer

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

Graph: the ray $y = 1$ for $x > 0$, the point O, the ray $y = -1$ for $x < 0$ (open ends at $x = 0$).

Grade - Low
Practice map, main concept: Sheet: C-I 5 | DPP: 2(15-16)
Also used in: none

Question 34
What is domain and range of a signum function?

Model Answer
$\operatorname{sgn} x$: domain R, range $\{-1, 0, 1\}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 35
Write properties of signum function.

Model Answer
$\operatorname{sgn} x = \dfrac{|x|}{x}$ $(x \neq 0)$; $x \operatorname{sgn} x = |x|$; $|x| \operatorname{sgn} x = x$; $\operatorname{sgn}(\operatorname{sgn} x) = \operatorname{sgn} x$; $\operatorname{sgn} f(x)$ depends only on the sign of $f(x)$.
Ex: $\operatorname{sgn}(x^2 - 3x + 4) = 1$ for all x (discriminant < 0).

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 37; C-I 20 | DPP: 3(12); 4(12)

Question 36
Define $y = [x]$.

Model Answer
$[x]$ = greatest integer $\leq x$: $[x] = n$ for $n \leq x < n+1$. Properties: $[x] \leq x < [x]+1$; $[x+m] = [x]+m$ $(m \in I)$; $[x] + [-x] = 0$ if $x \in I$, -1 otherwise; $[x] + [y] \leq [x + y] \leq [x] + [y] + 1$.
Ex: $[2.7] = 2$, $[-2.7] = -3$.

Grade - Medium
Practice map, main concept: Sheet: J-I 2; J-M 4 | DPP: 1(1); 2(13) | Question Bank: 69
Also used in: Sheet: J-I 28 | DPP: 2(8, 15) | Question Bank: 56

Question 37
What is domain and range of $y = [x]$?

Model Answer
$[x]$: domain R, range I (set of integers); graph is a staircase, closed dot at the left end of each step.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 38
How to define $y = [f(x)]$?

Model Answer
Split the domain at the points where $f(x)$ crosses an integer; on each part $[f(x)]$ is a constant.
Ex: $[2x]$ on $[1.2, 1.8]$: 2 for $x \in [1.2, 1.5)$, 3 for $x \in [1.5, 1.8]$.

Grade - Low
Practice map, main concept: DPP: 2(8)
Also used in: none

Question 39
How to solve equation $[f(x)] =$ constant?

Model Answer
$[f(x)] = k$ has solutions only if $k \in I$, and then $k \le f(x) < k + 1$. In mixed equations put $[x] = n$ (or $x = n + f$, $0 \le f < 1$), solve for each integer n and keep consistent values.
Ex: $[x] = 3 \Rightarrow x \in [3, 4)$; $[x] = \frac{3}{2}$ has no solution.

Grade - Medium
Practice map, main concept: Sheet: J-II 1; J-M 3; C-I 17; C-II 5 | Question Bank: 58, 62, 67
Also used in: none

Question 40
How to solve inequality $[f(x)]$ greater than/less than constant?

Model Answer
For $k \in I$: $[f] \ge k \iff f \ge k$; $[f] > k \iff f \ge k{+}1$; $[f] \le k \iff f < k{+}1$; $[f] < k \iff f < k$.
For non-integer k first replace by the nearest admissible integer.
Ex: $[x] < 7.6 \iff [x] \le 7 \iff x < 8$; $[x] > 5.6 \iff x \ge 6$.

Grade - Medium
Practice map, main concept: Sheet: J-I 7 | Question Bank: 56
Also used in: Sheet: J-M 1, 25

Question 41
Define $y = \{x\}$.

Model Answer
Fractional part: $\{x\} = x - [x]$; $\{x\} = x - n$ for $n \le x < n + 1$ (saw-tooth graph).
Ex: $\{2.1\} = 0.1$, $\{-3.7\} = 0.3$.

Grade - Medium
Practice map, main concept: DPP: 2(14)
Also used in: Sheet: J-II 1; C-I 14 | Question Bank: 62

Question 42
What is domain and range of $y = \{x\}$?

Model Answer
$\{x\}$: domain R, range $[0, 1)$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 2(9)

Question 43
How to define $y = \{f(x)\}$?

Model Answer
$\{f(x)\} = f(x) - [f(x)]$: on the part where $k \le f(x) < k+1$, $\{f(x)\} = f(x) - k$.
Ex: $\{2x\}$ on $[0, 1)$: $2x$ for $x \in [0, \frac{1}{2})$, $2x - 1$ for $x \in [\frac{1}{2}, 1)$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 44
Write properties of $y = \{x\}$.

Model Answer
$0 \le \{x\} < 1$; $\{[x]\} = [\{x\}] = 0$; $\{\{x\}\} = \{x\}$; $\{x + m\} = \{x\}$ ($m \in I$, period 1); $\{x\} + \{-x\} = 0$ if $x \in I$, 1 otherwise.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 45
What is the condition on n while expressing nth root of a?

Model Answer
$\sqrt[n]{a} = a^{1/n}$ with $n \in N$, $n \ge 2$. Even n: need $a \ge 0$; odd n: any real a. For $a^{p/q}$ (lowest terms): q even $\Rightarrow a \ge 0$; q odd $\Rightarrow a \in R$.

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-I 18; J-II 2

Question 46
What is the condition on base in case of a^x?

Model Answer
a^x is defined for every real x only if $a > 0$ ($a \ne 1$ for an exponential function). For $f(x)^{g(x)} = e^{g(x) \ln f(x)}$ take $f(x) > 0$.

Ex: domain of x^x is $x > 0$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 47
What is the condition to define ${}^nP_r$ and ${}^nC_r$?

Model Answer
${}^nP_r$, ${}^nC_r$: $n \in N$, $r \in W$ (whole number), $r \le n$.
Ex: ${}^{7-x}P_{x-3}$: $x \in \{3, 4, 5\}$, values $1, 3, 2$, so range $\{1, 2, 3\}$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 12 | DPP: 1(5)
Also used in: none

Question 48
How to find domain of $y = \max(f(x), g(x))$ or $y = \min(f(x), g(x))$?

Model Answer
Domain $= D_f \cap D_g$ (both must be defined).

Grade - Medium
Practice map, main concept: DPP: 1(2) | Question Bank: 45
Also used in: none

Question 49
How to define $y = \max(f(x), g(x))$ or $y = \min(f(x), g(x))$?

Model Answer
Draw both graphs and find where $f = g$; max follows the upper graph, min the lower graph on each interval.
Ex: $\max\{1 + x, 1 - x, 2\} = 1 - x$ $(x \leq -1)$, 2 $(-1 \leq x \leq 1)$, $1 + x$ $(x \geq 1)$.

Grade - Low
Practice map, main concept: Sheet: J-II 18
Also used in: none

Question 50
How to plot graph of $y = f(ax)$ from the graph of $y = f(x)$?

Model Answer
$a > 0$: divide every x-coordinate by a (compress horizontally if $a > 1$, stretch if $0 < a < 1$); y-values unchanged.
Ex: $\sin 2x$ has period π; $\sin \frac{x}{3}$ has period 6π.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 51
How to plot graph of $y = f(x + a)$ from the graph of $y = f(x)$?

Model Answer
Shift the graph a units to the left if $a > 0$, $|a|$ units to the right if $a < 0$.
Ex: $\sin\left(x + \frac{\pi}{6}\right)$ is $\sin x$ moved $\frac{\pi}{6}$ to the left. Symmetry about $x = a$ means $f(a + x) = f(a - x)$.

Grade - Low
Practice map, main concept: Sheet: J-I 13
Also used in: none

Question 52
Plot graph of $y = x^3$.

Model Answer
$y = x^3$: increasing on R, odd (symmetric about O), flat at O (point of inflection), below the x-axis for $x < 0$; domain and range R.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 53

Plot graph of $y = x + \dfrac{1}{x}$.

Model Answer

Odd; minimum 2 at $x = 1$ and maximum -2 at $x = -1$; asymptotes $x = 0$ and $y = x$; range $(-\infty, -2] \cup [2, \infty)$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 20; C-I 6

Question 54

Plot graph of $y = x - \dfrac{1}{x}$.

Model Answer

Odd; increasing on each side ($y' = 1 + \frac{1}{x^2} > 0$); zeros $x = \pm 1$; asymptotes $x = 0$ and $y = x$; range R.

Grade - Low

Practice map, main concept: none

Also used in: none

Range Based Questions

Question 55

How to find range when x can be expressed in terms of y?

Model Answer

Put $y = f(x)$, solve $x = g(y)$; the range is the set of y for which $g(y)$ is real and lies in the domain of f.

Ex: $y = \dfrac{2x+5}{3x-4} \Rightarrow x = \dfrac{4y+5}{3y-2}$, so range $R - \left\{\frac{2}{3}\right\}$.

Grade - Low

Practice map, main concept: Sheet: J-I 11 | DPP: 1(11); 2(3)

Also used in: none

Question 56

How to find range if function has been expressed in implicit form?

Model Answer

Write $y = f(x)$ as a quadratic in x; real x needs $D \geq 0$ (treat the case 'coefficient of $x^2 = 0$' separately and remove y-values given only by excluded x).

Ex: $y = \dfrac{x^2 - x + 1}{x^2 + x + 1} \Rightarrow (y-1)x^2 + (y+1)x + (y-1) = 0$, $D \geq 0 \Rightarrow y \in \left[\frac{1}{3}, 3\right]$.

Grade - Low

Practice map, main concept: DPP: 2(12)

Also used in: Sheet: C-II 12

Question 57
How to find range in case of $f(g(x))$ if $g(x)$ is monotonic function?

Model Answer
Find the set T of values of the inner expression; then range $= f(T)$. If the function applied on T is monotonic, the range lies between its values at the ends of T (order reversed if decreasing).
Ex: $e^{2\sin x}$: $2\sin x \in [-2, 2]$, range $[e^{-2}, e^{2}]$; $\ln(x^2 - 2x + 2)$: range $[0, \infty)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 8-10; J-II 27; J-M 9 | DPP: 2(1, 4-6, 9) | Question Bank: 43
Also used in: Sheet: J-II 30; C-I 10; C-II 17 | DPP: 4(8) | Question Bank: 40

Question 58
How to find range in case of $f(g(x))$ if $g(x)$ is non-monotonic function?

Model Answer
End values are not enough: include the greatest and least values attained inside (turning points). Use the graph of the function on the given set, or substitute $t = g(x)$ and find the range of t first.
Ex: $\sin x$ on $[0, 4]$: $\frac{\pi}{2} \in [0, 4]$, range $[\sin 4, 1]$.

Grade - Low
Practice map, main concept: Sheet: J-II 4
Also used in: Sheet: J-I 7 | DPP: 2(2, 6)

Question 59
How to find range of a non-uniform function?

Model Answer
Find the range of each piece on its own interval and take the union.
Ex: $f = x^2$ $(x < 0)$, $x + 1$ $(x \geq 0)$: $(0, \infty) \cup [1, \infty) = (0, \infty)$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 29 | Question Bank: 53

Equal/Identical Functions

Question 60
How to check whether two functions are identical?

Model Answer
f and g are identical iff (i) $D_f = D_g$ and (ii) $f(x) = g(x)$ for every x in that domain (co-domains equal when given).
Ex: $\ln e^x$ (domain R) and $e^{\ln x}$ (domain $(0, \infty)$) are not identical; $x \operatorname{sgn} x$ and $|x|$ are.

Grade - Low
Practice map, main concept: Sheet: J-II 5, 31; C-I 13
Also used in: none

Question 61
What is the relation between graphs of two identical functions?

Model Answer
Their graphs coincide exactly (same points, same gaps).
Ex: $\frac{x^2}{x}$ and x differ only at $x = 0$, so they are not identical.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Transformation of Graphs

Question 62
How to plot the graph of $y = f(x) + b$ from the graph of $y = f(x)$?

Model Answer
Shift the graph b units up if $b > 0$, $|b|$ units down if $b < 0$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 40
Also used in: none

Question 63
How to plot the graph of $y = -f(x)$ from the graph of $y = f(x)$?

Model Answer
Take the mirror image of the whole graph in the x-axis.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 64
How to plot the graph of $y = f(-x)$ from the graph of $y = f(x)$?

Model Answer
Take the mirror image of the whole graph in the y-axis.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 65
How to plot the graph of $y = f(|x|)$ from the graph of $y = f(x)$?

Model Answer
Delete the part left of the y-axis; replace it by the mirror image of the right part ($x \geq 0$) in the y-axis. $f(|x|)$ is even.
Ex: $e^{|x|}$; $f(|x|) = k$ has 4 roots when $y = k$ cuts the right part twice for $x > 0$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 1; J-II 28, 30; C-I 14; C-II 19
Also used in: Sheet: J-II 26, 40; C-II 21 | DPP: 2(11)

Question 66
How to plot the graph of $y = f(-|x|)$ from the graph of $y = f(x)$?

Model Answer
Delete the part right of the y-axis; replace it by the mirror image of the left part ($x \leq 0$) in the y-axis.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 67
How to plot the graph of $|y| = f(x)$ from the graph of $y = f(x)$?

Model Answer
Delete the part below the x-axis; add the mirror image of the remaining part in the x-axis (no points where $f(x) < 0$). It is not a function.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 40

Question 68
How to plot the graph of $y = \dfrac{\text{linear}}{\text{linear}}$?

Model Answer
Write $y = \dfrac{ax+b}{cx+d}$ as $y - \dfrac{a}{c} = \dfrac{k}{x+\frac{d}{c}}$: a rectangular hyperbola with centre $\left(-\dfrac{d}{c}, \dfrac{a}{c}\right)$ and asymptotes $x = -\dfrac{d}{c}$, $y = \dfrac{a}{c}$; branches in the 1st/3rd quadrants about the centre if $k > 0$, 2nd/4th if $k < 0$.
Ex: $y = \dfrac{2x+1}{x+4} = 2 - \dfrac{7}{x+4}$: centre $(-4, 2)$, $k = -7 < 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Important Definitions

Question 69
Define identity function.

Model Answer
$I_A : A \to A$, $I_A(x) = x$ for all $x \in A$; it is a bijection (graph $y = x$).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 70
Define constant function.

Model Answer
$f(x) = c$ for all $x \in A$; range is the singleton $\{c\}$; many-one (if A has two or more elements); periodic with no fundamental period.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 71
How to check whether a function is bounded function?

Model Answer
f is bounded if $|f(x)| \leq M$ for some finite M, i.e. its range does not extend to $+\infty$ or $-\infty$.
Ex: $\sin x$, $\dfrac{1}{1+x^2}$ bounded; $x + \dfrac{1}{x}$ not.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 72
How to check whether a function is homogeneous function and find its degree?

Model Answer
$f(x, y)$ is homogeneous of degree n if $f(tx, ty) = t^n f(x, y)$ (every term has the same degree).
Ex: $\dfrac{x^2+y^2}{x+5y}$: degree 1; $x \ln \frac{x}{y} + y$: degree 1; $\sin(xy)$: not homogeneous.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 73
How to convert implicit form of a function to its explicit form?

Model Answer
Solve the relation for y to get $y = f(x)$ and state the domain on which the solution is real.
Ex: $2^x + 2^y = 4 \Rightarrow y = \log_2(4 - 2^x)$, $x < 2$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 6
Also used in: none

One-One and Many-One Function

Question 74
What do you mean by one-one and many-one function?

Model Answer
One-one (injective): $f(a) = f(b) \Rightarrow a = b$ (distinct elements, distinct images). Many-one: some $x_1 \neq x_2$ with $f(x_1) = f(x_2)$.
Ex: $2x + 3$ is one-one; x^2 on R is many-one ($f(-1) = f(1)$).

Grade - **High**
Practice map, main concept: Sheet: J-I 15; J-II 7; J-M 13-14, 24; C-I 2, 12; C-II 1 | DPP: 3(6-8, 12) | Question Bank: 52
Also used in: Sheet: J-II 34; J-A 13; C-I 17 | DPP: 3(4, 9); 5(12) | Question Bank: 51, 55

Question 75
How to determine whether a function will be one-one or many-one from its graph?

Model Answer
Horizontal line test: one-one iff every horizontal line cuts the graph at most once; if some line cuts it twice or more, many-one.

Grade - Low
Practice map, main concept: Sheet: J-II 32 | DPP: 3(9)
Also used in: none

Question 76
How to determine whether a function will be one-one or many-one with the help of derivative approach?

Model Answer
For a continuous f on an interval: $f'(x) \geq 0$ throughout (or ≤ 0), equality only at isolated points $\Rightarrow$ one-one; f' takes both signs $\Rightarrow$ many-one.
Ex: $x + \sin x$: $f' = 1 + \cos x \geq 0$, one-one; $3\cos x - 2x$: $f' \in [-5, 1]$, many-one.

Grade - Medium
Practice map, main concept: Sheet: J-A 1, 5; C-I 4 | DPP: 3(1-2, 5) | Question Bank: 46
Also used in: Sheet: J-I 14; J-II 7, 23; J-A 3

Question 77
How to determine whether a cubic polynomial will be one-one or many-one?

Model Answer
$f = ax^3 + bx^2 + cx + d$ is one-one iff $f' = 3ax^2 + 2bx + c$ never changes sign, i.e. its discriminant ≤ 0: $b^2 \leq 3ac$; otherwise many-one. (Always onto R.)
Ex: $x^3 - 6x^2 + 12x - 7$: $36 \leq 36$, $f' = 3(x-2)^2 \geq 0$, one-one.

Grade - Low
Practice map, main concept: Sheet: J-I 14, 17; J-II 23-24; C-I 1
Also used in: Sheet: J-II 25

Question 78
What are the standard examples of one-one function?

Model Answer
$ax + b$ ($a \neq 0$); $\dfrac{ax+b}{cx+d}$ with no common factor ($ad \neq bc$); a^x; $\log_a x$; x^3; $x|x|$; any strictly monotonic function.

Grade - Low
Practice map, main concept: DPP: 3(10)
Also used in: none

Question 79
What are the standard examples of many-one function?

Model Answer
Even degree polynomials; $\dfrac{\text{quadratic}}{\text{linear}}$, $\dfrac{\text{linear}}{\text{quadratic}}$, $\dfrac{\text{quadratic}}{\text{quadratic}}$ (no common factor); all trigonometric functions on their whole domain; $|x|$, $[x]$, $\{x\}$, $\operatorname{sgn} x$; even and periodic functions.

Grade - Low
Practice map, main concept: DPP: 3(3)
Also used in: none

Onto and Into Function

Question 80

What do you mean by onto and into function?

Model Answer

Onto (surjective): every element of the co-domain has a pre-image, i.e. range = co-domain. Into: some element of the co-domain has no pre-image (range $\subsetneq$ co-domain).

Ex: $|x| : R \to R$ is into; $|x| : R \to [0, \infty)$ is onto.

Grade - High

Practice map, main concept: Sheet: J-I 16; J-II 34; C-I 16, 18; C-II 12 | DPP: 3(4, 11) | Question Bank: 49

Also used in: Sheet: J-I 15; J-II 7; J-M 13-14, 24; J-A 1, 3, 13; C-I 2 | DPP: 3(1-3, 6-8, 10, 12) | Question Bank: 46, 51-52, 55

Question 81

How to check whether a polynomial function will be onto or into function?

Model Answer

$f : R \to R$ polynomial: odd degree $\Rightarrow$ always onto (no general rule for one-one); even degree $\Rightarrow$ neither onto nor one-one.

Ex: $x^3 + 5x + 1$ is onto and one-one; $x^4 + 2x^3 - x^2 + 1$ is neither.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 17; J-II 24; J-A 5; C-I 1, 4, 21 | DPP: 3(5)

Question 82

What do you mean by invertible functions?

Model Answer

$f : A \to B$ is invertible iff it is one-one and onto (bijective); such functions are also called non-singular or biuniform.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 83

What do you mean by injective mapping?

Model Answer

Injective mapping = one-one function: $a \neq b \Rightarrow f(a) \neq f(b)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 84

What do you mean by surjective mapping?

Model Answer

Surjective mapping = onto function: for every $b \in B$ there is $a \in A$ with $f(a) = b$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 85
What do you mean by bijective mapping?

Model Answer
Bijective = one-one and onto. A function is one of: one-one onto, one-one into, many-one onto, many-one into.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 86
How to check whether a function defined separately for rational and irrational numbers will be one-one/many-one?

Model Answer
Check (i) each rule is one-one on its own set and (ii) no image of a rational equals the image of an irrational; onto: images of both sets together fill the co-domain.
Ex: $f(x) = x$ $(x \in Q)$, $-x$ $(x \notin Q)$ is one-one and onto on R.

Grade - Low
Practice map, main concept: Sheet: C-I 11
Also used in: none

Permutation Based Problems

Question 87
How to find number of functions that can be defined from A to B if $n(A) = n(B)$?

Model Answer
$n(A) = n(B) = n$: total functions n^n (each element of A has n choices). With conditions, count choices element by element.
Ex: $n = 4$: $4^4 = 256$.

Grade - Medium
Practice map, main concept: Sheet: J-M 7; J-A 11
Also used in: none

Question 88
How to find number of one-one/many-one functions that can be defined from A to B if $n(A) = n(B)$?

Model Answer
One-one = $n!$ (all bijections); many-one = $n^n - n!$.
Ex: $n = 4$: one-one 24, many-one 232.

Grade - Medium
Practice map, main concept: Sheet: J-M 6, 23 | Question Bank: 42
Also used in: none

Question 89
How to find number of onto/into functions that can be defined from A to B if $n(A) = n(B)$?

Model Answer
Onto $= n!$ (onto $\iff$ one-one here); into $= n^n - n!$.
Ex: $n = 4$: onto 24, into 232.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 90
How to find number of functions that can be defined from A to B if $n(A) > n(B)$?

Model Answer
$n(A) = m > n(B) = n$: total n^m.
Ex: $m = 5$, $n = 4$: $4^5 = 1024$.

Grade - **Medium**
Practice map, main concept: Sheet: C-II 14 | Question Bank: 60
Also used in: none

Question 91
How to find number of one-one/many-one functions that can be defined from A to B if $n(A) > n(B)$?

Model Answer
One-one: none (0); many-one: all n^m.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 92
How to find number of onto/into functions that can be defined from A to B if $n(A) > n(B)$?

Model Answer

$$\text{Onto} = \sum_{r=0}^{n} (-1)^r \, {}^nC_r \, (n-r)^m, \qquad \text{Into} = n^m - \text{Onto}$$

Or: divide A into n non-empty groups and arrange the groups on B ($\times n!$).
Ex: $m = 5$, $n = 4$: $1024 - 972 + 192 - 4 = 240$ onto, 784 into.

Grade - **Medium**
Practice map, main concept: Sheet: J-M 19; C-I 15 | Question Bank: 66
Also used in: Sheet: J-A 4

Question 93
How to find number of functions that can be defined from A to B if $n(A) < n(B)$?

Model Answer
$n(A) = m < n(B) = n$: total n^m; with a condition, count the choices of the restricted elements first.
Ex: $m = 4$, $n = 5$: $5^4 = 625$.

Grade - **Medium**
Practice map, main concept: Sheet: J-M 22, 28; C-II 13 | Question Bank: 65, 68
Also used in: Question Bank: 59

Question 94
How to find number of one-one/many-one functions that can be defined from A to B if $n(A) < n(B)$?

Model Answer
One-one $= {}^nP_m = \dfrac{n!}{(n-m)!}$; many-one $= n^m - {}^nP_m$.
Ex: $m = 4$, $n = 5$: one-one 120, many-one 505.

Grade - Medium
Practice map, main concept: Sheet: J-M 15; J-A 4 | Question Bank: 41, 50, 64
Also used in: none

Question 95
How to find number of onto/into functions that can be defined from A to B if $n(A) < n(B)$?

Model Answer
Onto: none (0); into: all n^m.

Grade - Low
Practice map, main concept: none
Also used in: none

Functional Equation

Question 96
How to solve functional equation of type 1?

Model Answer
$f(g(x)) = h(x)$ given: put $g(x) = t$, express x in t, get $f(t)$, then replace t by x. Also try special values ($x = 0$, $y = 0$, $x = y$) in the given relation.
Ex: $f(2x+1) = x + 4 \Rightarrow f(t) = \dfrac{t-1}{2} + 4 \Rightarrow f(x) = \dfrac{x+7}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 20-21; J-II 8, 10; J-A 2; C-II 4, 9, 11 | Question Bank: 59
Also used in: Sheet: J-M 20-21

Question 97
How to solve functional equation of type 2 and when to apply?

Model Answer
Apply when the relation has $f(x)$ and $f(g(x))$ with $g(g(x)) = x$ (e.g. $\frac{1}{x}$, $1-x$, $-x$): replace x by $g(x)$ to get a second equation and solve the two linear equations for $f(x)$.
Ex: $2f(x) + f(1-x) = 2x - 1$ and $2f(1-x) + f(x) = 1 - 2x \Rightarrow f(x) = 2x - 1$.

Grade - Medium
Practice map, main concept: Sheet: J-I 23; C-II 8 | Question Bank: 38
Also used in: none

Question 98

How to solve functional equation of type 3 and when to apply?

Model Answer

Apply when f is an odd (or even) expression plus a constant: $f(x)+f(-x)=2c$ (or $f(x)-f(-x)=$ known); find $f(-a)$ from $f(a)$ without the unknown coefficients.

Ex: $f(x)=ax^5+bx^3+cx+d\sin x+4$, $f(3)=7 \Rightarrow f(-3)=8-7=1$.

Grade - Low

Practice map, main concept: Sheet: C-II 3

Also used in: Sheet: J-II 8

Question 99

How to solve the standard functional equations $f(x+y)=f(x)+f(y)$, $f(x+y)=f(x)f(y)$ and $f(xy)=f(x)+f(y)$?

Model Answer

$f(x+y)=f(x)+f(y)$: $f(0)=0$, $f(nx)=nf(x)$, so $f(x)=kx$ with $k=f(1)$ (on N: $f(n)=nf(1)$).
$f(x+y)=f(x)f(y)$ $(f\not\equiv 0)$: $f(0)=1$, $f(x)=a^x$ with $a=f(1)$ (on N: $f(n)=f(1)^n$).
$f(xy)=f(x)+f(y)$: $f(1)=0$, $f(x)=k\log x$.

Ex: $f(x+y)=f(x)+f(y)$, $f(1)=7 \Rightarrow \displaystyle\sum_{r=1}^{n} f(r)=\frac{7n(n+1)}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 22, 24; J-M 8, 11, 20; J-A 9, 14 | Question Bank: 63

Also used in: none

Question 100

How to find a polynomial (or its value) when its values at several points are given?

Model Answer

If $f(i)=g(i)$ for $i=1,\ldots,n$, then $h(x)=f(x)-g(x)$ has roots $1,\ldots,n$: $h(x)=k(x-1)(x-2)\cdots(x-n)$ (times a further factor if the degree is higher); fix k from the leading coefficient or one more value.

Ex: $P(x)=x^4+\ldots$, $P(0)=6$, $P(1)=7$, $P(2)=8$, $P(3)=9 \Rightarrow P(x)-x-6=x(x-1)(x-2)(x-3) \Rightarrow P(4)=34$.

Grade - Medium

Practice map, main concept: Sheet: J-II 19; J-M 10; C-I 21; C-II 2

Also used in: none

Question 101

How to evaluate sums like $\sum f\left(\frac{k}{n}\right)$ when $f(x)+f(1-x)$ is constant?

Model Answer

Check $f(x)+f(1-x)=c$; pair the terms $\frac{k}{n}$ and $\frac{n-k}{n}$: $\displaystyle\sum_{k=1}^{n-1} f\left(\frac{k}{n}\right)=\frac{(n-1)c}{2}$ (add any extra term separately).

Ex: $f(x)=\dfrac{4^x}{4^x+2}$: $f(x)+f(1-x)=1$, so $\displaystyle\sum_{k=1}^{39} f\left(\frac{k}{40}\right)=\frac{39}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 21; J-M 17; J-A 6 | Question Bank: 44, 47

Also used in: none

Question 102
Which polynomials satisfy $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$?

Model Answer
Only $f(x) = 1 + x^n$ and $f(x) = 1 - x^n$ $(n \in N)$; one given value fixes the sign and n.
Ex: $f(3) = 28 \Rightarrow 1 + 3^n = 28 \Rightarrow n = 3$, $f(x) = 1 + x^3$, $f(4) = 65$.

Grade - Medium
Practice map, main concept: Question Bank: 48
Also used in: none

Composite of Two Functions

Question 103
How to define composite of two functions if their functional rule is given?

Model Answer
$(g \circ f)(x) = g(f(x))$; domain $= \{x \in D_f : f(x) \in D_g\}$ (range of f must lie in domain of g). Not commutative, but associative.
Ex: $f = \sqrt{x-4}$, $g = x^2$: $f \circ g = \sqrt{x^2 - 4}$, domain $(-\infty, -2] \cup [2, \infty)$; $g \circ f = x - 4$, domain $[4, \infty)$.

Grade - High
Practice map, main concept: Sheet: J-I 26; J-II 9, 39; J-M 21; C-I 10 | DPP: 4(1, 3) | Question Bank: 40, 57
Also used in: Sheet: J-I 3, 25; J-II 10, 15, 22; J-M 2, 13, 16, 18; C-I 5; C-II 24 | Question Bank: 54

Question 104
How to define composite of two functions if functions are given in arrow diagram form?

Model Answer
Follow the arrows $x \to f(x) \to g(f(x))$ for every element; $g \circ f$ exists only if every $f(x)$ is in the domain of g.
Ex: $f = \{(1, a), (2, b)\}$, $g = \{(a, p), (b, p)\} \Rightarrow g \circ f = \{(1, p), (2, p)\}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 18
Also used in: none

Question 105
What is the condition on f and g if $f(g(x))$ is one-one?

Model Answer
$f \circ g$ one-one $\Rightarrow g$ is one-one (f need be one-one only on the range of g). f, g both one-one $\Rightarrow f \circ g$ one-one; g many-one $\Rightarrow f \circ g$ many-one.
Ex: $f = x^2$, $g = e^x$: $f \circ g = e^{2x}$ is one-one though f is not.

Grade - Medium
Practice map, main concept: Sheet: J-M 5; J-A 13; C-I 19
Also used in: none

Question 106
What is the condition on f and g if $f(g(x))$ is onto?

Model Answer
$f \circ g$ onto $\Rightarrow$ f is onto (g need not be). f, g both onto $\Rightarrow$ $f \circ g$ onto; f into $\Rightarrow$ $f \circ g$ into. Composite of two bijections is a bijection.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 5; J-A 13

Question 107
How to solve questions of composite function if higher number of times composite has to be taken?

Model Answer
Compute $f \circ f$, $f \circ f \circ f, \ldots$ until a pattern or a cycle (f^k = identity) appears; then reduce n modulo k. For linear $f = px + q$: $f^n = p^n x + q(1 + p + \cdots + p^{n-1})$.
Ex: $f = \dfrac{x-1}{x+1}$: $f^2 = -\dfrac{1}{x}$, $f^4(x) = x$, so f^n repeats with period 4.

Grade - Medium
Practice map, main concept: Sheet: J-M 12, 16 | DPP: 4(2) | Question Bank: 54, 61
Also used in: none

Composite of Non-uniform Functions

Question 108
How to find composite of two non-uniform functions?

Model Answer
Write $f(g(x))$ by the pieces of f with $g(x)$ as the variable; from the graph of g find the x-intervals where $g(x)$ lies in each piece of D_f (drop x where $g(x) \notin D_f$).
Ex: $f = x+1$ $(x \le 1)$, $2x+1$ $(1 < x \le 2)$; $g = x^2$ $(-1 \le x \le 2)$, $x+2$ $(2 < x \le 3)$: $f(g(x)) = x^2+1$ $(-1 \le x \le 1)$, $2x^2+1$ $(1 < x \le \sqrt{2})$.

Grade - Medium
Practice map, main concept: Sheet: J-II 11; C-II 10, 17 | DPP: 4(11) | Question Bank: 51, 53, 55
Also used in: Sheet: C-I 5

Odd and Even Functions

Question 109
Define odd and even functions.

Model Answer
Even: $f(-x) = f(x)$; odd: $f(-x) = -f(x)$ for all x in the domain. A function may be neither.
Ex: $\cos x$, $x^3 \sin x$ even; $\ln \dfrac{1-x}{1+x}$, $\ln(x + \sqrt{1+x^2})$ odd; $\sin x + \cos x$ neither.

Grade - Medium
Practice map, main concept: Sheet: J-I 27-28; J-II 13, 35; J-A 3; C-II 7 | DPP: 4(4-6, 12)
Also used in: Sheet: J-II 16, 36; J-A 5; C-I 12, 16 | DPP: 5(12)

Question 110
What is the condition on domain for odd/even function?

Model Answer
The domain must be symmetric about 0 ($x \in D \Rightarrow -x \in D$); otherwise the function is neither. The only function that is both even and odd on R is $f(x) = 0$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 111
What can you say about graphs of odd/even function?

Model Answer
Even: symmetric about the y-axis (hence many-one). Odd: symmetric about the origin; if $0 \in D$ then $f(0) = 0$.

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-I 13

Question 112
What is the nature of derivative of odd functions?

Model Answer
Derivative of an odd function is even.
Ex: $(\sin x)' = \cos x$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 113
What is the nature of derivative of even functions?

Model Answer
Derivative of an even function is odd.
Ex: $(x^2)' = 2x$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 114
How to express a function as a sum of even and odd functions?

Model Answer

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}$$

Ex: f odd, g even, $f - g = x^2 + 5x + 7 \Rightarrow -g = x^2 + 7$, $g(2) = -11$.

Grade - **Low**
Practice map, main concept: DPP: 4(7)
Also used in: none

Question 115
What is nature of $f+g$, $f-g$, $f.g$, f/g if f and g are both even/both odd/one is even and other is odd?

Model Answer
Both even: all even. Both odd: $f \pm g$ odd, fg and f/g even. One even, one odd (non-zero): $f \pm g$ neither, fg and f/g odd.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 116
What is the nature of $f(g(x))$, $g(f(x))$ if f and g are both even/both odd/one is even and other is odd?

Model Answer
If either function is even, both composites are even; if both are odd, both composites are odd.
Ex: $\cos(\sin x)$ even; $\sin(\tan x)$ odd.

Grade - Low
Practice map, main concept: Sheet: C-I 20
Also used in: none

Inverse of a Function

Question 117
What is the condition for a function to be invertible?

Model Answer
f must be one-one and onto (bijective). A many-one function can be inverted only after restricting its domain (and taking co-domain = range).

Grade - Medium
Practice map, main concept: Sheet: J-II 25; C-II 18
Also used in: Sheet: J-M 5

Question 118
Define inverse of a function.

Model Answer
For a bijection $f : A \to B$, $f^{-1} : B \to A$ with $f^{-1}(y) = x \iff f(x) = y$. Domain of f^{-1} = range of f; range of f^{-1} = domain of f; the inverse is unique and bijective.
Ex: $f(x) = x^3 - 1$ on $\{0, 1, 2, 3\}$: domain of f^{-1} is $\{-1, 0, 7, 26\}$.

Grade - Low
Practice map, main concept: DPP: 5(1)
Also used in: none

Question 119
How to find inverse of a function if functional rule is given?

Model Answer
Put $y = f(x)$, solve for x in terms of y, interchange x and y; choose the branch that matches the range of f. To find $f^{-1}(a)$ just solve $f(x) = a$.
Ex: $f = \dfrac{x+5}{3x-4} \Rightarrow f^{-1}(x) = \dfrac{4x+5}{3x-1}$; $f : [2, \infty) \to [1, \infty)$, $x^2 - 4x + 5 \Rightarrow f^{-1} = 2 + \sqrt{x-1}$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 29; J-II 14; C-I 6, 8; C-II 20, 24 | DPP: 4(10); 5(2, 5, 13)
Also used in: Sheet: J-II 12; C-I 3, 18; C-II 4

Question 120
How to find inverse of a function if function is given as a set/arrow diagram form?

Model Answer
Check that f is a bijection, then reverse every pair $(a, b) \to (b, a)$ (reverse every arrow).
Ex: $f = \{(1, a), (2, b), (3, c)\} \Rightarrow f^{-1} = \{(a, 1), (b, 2), (c, 3)\}$.

Grade - **Low**
Practice map, main concept: DPP: 5(3)
Also used in: none

Question 121
What is inverse of inverse of a function?

Model Answer

$$(f^{-1})^{-1} = f$$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 122
What is composite of f and g if f and g are inverse of each other?

Model Answer
$f \circ g = I_B$ and $g \circ f = I_A$: $f(g(x)) = x$ and $g(f(x)) = x$ on the respective domains.
Ex: f, g linear and inverse to each other, $f(0) = 4$, $g(5) = 17 \Rightarrow f(17) = 5$, so $f(x) = \dfrac{x}{17} + 4$ and $f(136) = 12$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 12
Also used in: none

Question 123
What is the relation between graphs of $f(x)$ and its inverse?

Model Answer
They are mirror images in the line $y = x$; (a, b) on $f \iff (b, a)$ on f^{-1}. For increasing continuous f, f and f^{-1} meet only on $y = x$: solve $f(x) = x$.
Ex: $f = (x+1)^2 - 1$, $x \geq -1$: $f(x) = f^{-1}(x) \iff x^2 + x = 0 \iff x \in \{0, -1\}$.

Grade - **Low**
Practice map, main concept: Sheet: C-I 3, 9
Also used in: none

Question 124
How to find inverse of composite function?

Model Answer

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Ex: $f \circ g(x) = 2x^3 + 7$: invert directly, $(f \circ g)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 125
What is the condition if $\frac{\text{linear}}{\text{linear}}$ is inverse of itself?

Model Answer
$f(x) = \frac{ax+b}{cx+d}$ $(ad \neq bc)$ satisfies $f \circ f = I$ iff $a + d = 0$ (or $f(x) = x$). Its graph is symmetric about $y = x$.
Ex: $\frac{2x+3}{x-2}$ and $\frac{1-x}{1+x}$ are their own inverses.

Grade - Medium
Practice map, main concept: Sheet: J-I 25; J-M 2; C-I 7 | DPP: 5(4)
Also used in: none

Periodic Function

Question 126
Define periodic function.

Model Answer
f is periodic if there is a fixed $T > 0$ with $f(x + T) = f(x)$ for all $x \in D$; the least such T is the fundamental period. Periodic functions are many-one.
Ex: $\{x\}$ has period 1; $[x]$, $\sin x^2$, $\sin^{-1} x$ are aperiodic.

Grade - Low
Practice map, main concept: Sheet: J-II 16, 33 | DPP: 5(12)
Also used in: Sheet: J-II 34

Question 127
What is the condition on domain of a periodic function?

Model Answer
The domain must repeat with period T $(x \in D \Rightarrow x + T \in D)$; a function with a bounded domain is aperiodic.
Ex: $\sin^{-1} x$ (domain $[-1, 1]$) is aperiodic.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 33

Question 128
What is the condition on points of discontinuity or points where function is not defined for a periodic function?

Model Answer
These points must also repeat periodically; include their spacing in the LCM.
Ex: $\sin^2 \frac{2x}{3}$, $x \neq n\pi$: LCM of $\frac{3\pi}{2}$ and π gives period 3π.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 129
What are the periods of all trigonometric functions?

Model Answer
$\sin, \cos, \sec, \operatorname{cosec}$: 2π; $\tan, \cot$: π; their moduli: π; $\sin^n x, \cos^n x, \sec^n x, \operatorname{cosec}^n x$: π (n even), 2π (n odd); $\tan^n x, \cot^n x$: π.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 130
What are the periods of other standard functions?

Model Answer
$\{x\} = x - [x]$: 1; $|\sin x| + |\cos x|$, $\sin^4 x + \cos^4 x$: $\frac{\pi}{2}$; constant function: periodic, no fundamental period; $[x]$, $x + \sin x$, polynomials: aperiodic.
Ex: $[x] + [2x] + [3x] - 6x + 7 = 7 - \{x\} - \{2x\} - \{3x\}$: period 1.

Grade - **Low**
Practice map, main concept: Sheet: J-I 31; J-II 36 | DPP: 5(9, 11)
Also used in: Sheet: J-II 17; C-II 15 | DPP: 5(7)

Question 131
What will be period of $af(x)$, $f(x) + a$, $\dfrac{1}{f(x)}$ if period of $f(x)$ is T?

Model Answer
All have the same period T (also $f(x + c)$).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 132
What will be period of $f(ax + b)$ if period of $f(x)$ is T?

Model Answer
Period of $f(ax + b) = \dfrac{T}{|a|}$
Ex: $\sin^4 \frac{x}{7}$: $\dfrac{\pi}{1/7} = 7\pi$; $\cos(3x + 1)$: $\dfrac{2\pi}{3}$.

Grade - **Low**
Practice map, main concept: DPP: 5(8)
Also used in: none

Question 133
What will be period of $f+g$, $f-g$, fg, f/g if periods of f and g are T_1 and T_2?

Model Answer
Period = LCM(T_1, T_2) (it exists only if T_1/T_2 is rational); the true period may be smaller, so check.
Ex: $\sin^2 \frac{2x}{3} + \cos^2 \frac{3x}{5}$: LCM $\left(\frac{3\pi}{2}, \frac{5\pi}{3}\right) = 15\pi$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 17, 37; C-II 15, 22 | DPP: 5(7, 14)
Also used in: Sheet: J-I 31 | DPP: 5(9)

Question 134
What to keep in mind after taking LCM of periods while computing period?

Model Answer
Check whether a smaller number works, especially half of the LCM: test $f\left(x + \frac{L}{2}\right) = f(x)$. LCM of a rational and an irrational period does not exist (aperiodic).
Ex: $|\sin x| + |\cos x|$: LCM π, but period $\frac{\pi}{2}$; $\sin x + \{x\}$ is aperiodic.

Grade - **Low**
Practice map, main concept: DPP: 5(10)
Also used in: none

Question 135
How to find LCM of two or more numbers?

Model Answer

$$\text{LCM}\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{LCM}(a, c, e)}{\text{HCF}(b, d, f)}$$

Irrational multiples of the same number ($k\pi$): take LCM of the coefficients. Ex: LCM $\left(\frac{2}{3}, \frac{4}{5}\right) = \frac{4}{1} = 4$.

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 5(10)

Question 136
How to find period of a composite function?

Model Answer
If g is periodic with period T, $f \circ g$ is periodic with period T or a divisor of it: test $T/2$ etc. If only f is periodic, $f \circ g$ may not be periodic.
Ex: $\cos(\cos x)$, $\sec(\sin x)$: π; $\sin(\cos x)$: 2π; $\sin\sqrt{x}$: aperiodic.

Grade - **Low**
Practice map, main concept: Sheet: J-I 30; J-II 15
Also used in: none

Question 137
How to find period if functional equation is given?

Model Answer
Replace x by $x + a$ (repeatedly) and combine the equations until $f(x + T) = f(x)$ appears.
Ex: $f(x + 1) + f(x + 3) = 2$ and $f(x + 3) + f(x + 5) = 2 \Rightarrow f(x + 1) = f(x + 5)$: period 4. $f(x - 3) = f(x + 6)$: period 9.

Grade - Low
Practice map, main concept: Sheet: J-II 38 | DPP: 5(6)
Also used in: none

Miscellaneous

Question 138
How to solve equation if $f(x) = k$ has a solution?

Model Answer
$f(x) = k$ has a solution iff k lies in the range of f: find the range and put k in it.
Ex: $\sin x + \cos x = k$ is solvable iff $-\sqrt{2} \leq k \leq \sqrt{2}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 139
How to solve equation if $f(x) = k$ has exactly n solutions?

Model Answer
Draw $y = f(x)$ (use transformations) and the horizontal line $y = k$; choose k so that the line cuts the graph exactly n times. For $f(f(x)) = k$ first solve $f(t) = k$, then count solutions of $f(x) = t$ for each root t.
Ex: $|x^2 - 4| = k$ has exactly 4 solutions for $0 < k < 4$.

Grade - Medium
Practice map, main concept: Sheet: J-II 22, 29; C-II 21 | DPP: 2(11)
Also used in: Question Bank: 45